

Sanjoli Narang

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Education

- Ongoing **Massachusetts Institute of Technology (MIT)** CGPA: 5.0/5.0
Ph.D. , Electrical Engineering and Computer Science
Thesis Advisor: Manya Ghobadi
- 2023-24 **Massachusetts Institute of Technology (MIT)** CGPA: 5.0/5.0
Master of Science , Electrical Engineering and Computer Science
Thesis Advisor: Manya Ghobadi
- 2017-21 **Indian Institute Of Technology (IIT) Bombay** CGPA: 9.78/10.0
B.Tech. , Electrical Engineering and minor in Computer Science
Ramesh Chandra Sinha Academic Excellence Award
- 2016-17 **D.A.V Public School, Thermal Panipat** Aggregate: 97.4%
Senior Secondary education with major in science and mathematics

Research Interests

Computer Networks, Systems for Machine Learning, Computer Architecture, Datacenters, Applied Mathematics, Compute Express Link (CXL), Memory Disaggregated Systems, Space Technology, Autonomous Systems

Publications

- 2026 **Bring Your Own Objective: Inter-operability of Network Objectives in Datacenters**
arXiv Preprint
Sanjoli Narang, Anup Agarwal, Venkat Arun and Manya Ghobadi (MIT)
- 2025 **MLCC: A Congestion Control Technique to Accelerate ML Training**
Under Submission
Sanjoli Narang, Anton Zabreyko*, Sudarsanan Rajasekaran and Manya Ghobadi (MIT)*
- 2025 **Development of a Compton Imager Setup**
Accepted at Journal of Astrophysics and Astronomy (JOAA)
Anuraag Arya et al. (IIT Bombay)
- 2024 **MLTCP: A Distributed Technique to Approximate Centralized Flow Scheduling For Machine Learning**
Hotnets-2024
Sudarsanan Rajasekaran, Sanjoli Narang, Anton Zabreyko*, and Manya Ghobadi (MIT)*
- 2024 **Science with the daksha high energy transients mission**
Experimental Astronomy 57, 23
Varun Bhalerao et al. (IIT Bombay)
- 2024 **Daksha: on alert for high energy transients**
Experimental Astronomy 57, 24
Varun Bhalerao et al. (IIT Bombay)

- 2022 **Field-programmable gate array (FPGA) based programmable digital emulator of vibratory microelectromechanical systems (MEMS) gyroscopes**
Review of Scientific Instruments, 93(3):035003, 03 2022
Sanjoli Narang and Siddharth Tallur (IIT Bombay)
- 2021 **A reconfigurable hardware emulator of mems gyroscopes with built-in error source models**
IEEE conference EFTF/IFCS -21
Sanjoli Narang and Siddharth Tallur (IIT Bombay)
- 2021 **Modelling of underwater acoustic networks for source localization in arbitrary bounded reservoirs**
IEEE conference EFTF/IFCS -21
Sanjoli Narang, Aditya Harakare, Nayan Barhate, Andrews Varghese, Aayush Shrivastava, and Ayushi Gupta (IIT Bombay)

Awards and Honors

- * MIT Jacobs Presidential Fellowship Award
- * Ramesh Chandra Sinha Academic Excellence Award as the best performing female of IIT Bombay
- * IEEE EFTF-IFCS-21 Best Student Paper Finalist
- * Undergraduate Research Excellence Award (twice; URA-01, URA-02), IIT Bombay
- * Young Researcher's Prize(to AUV-IITB) by IEEE Ocean Engineering Society UT-21
- * Desai Sethi scholarship as top performing female student of IIT Bombay
- * Institute Academic Prize, IIT Bombay
- * Department Rank 6 in Electrical Engineering major in IIT Bombay
- * All India Rank 260 in JEE Advanced out of 0.1 million candidates
- * All India Rank 177 in JEE Mains out of 1.2 million candidates
- * All India Rank 2 in All India Engineering Aptitude Test
- * State Academic Topper of Haryana in CBSE high school exams
- * All India Rank 81 in Akash National Talent Hunt Exam (ANTHE-2015)
- * Akash ANTHE-2015 JEE-batch fellowship

Work Experience

- 2021-23 **Google India Pvt. Ltd. Silicon Digital Design Engineer**
Micro-architecture and design of the interrupts and debug-trace hardware stack of system-on-chips
- 2020 **Tyndall National Institute, Ireland Research Assistant with Dr. Cleitus Antony**
Theory and analysis of optical sensing using Guided Acoustic Waves Brillouin Scattering
- 2020 **Metal Power Analytical Pvt. Limited Hardware Engineer**
Developed hardware model of a generalised FPGA card for data acquisition in spectrometers
- 2020 **Larsen & Toubro Defence, IIT Bombay Overall Design Lead**
Design of remotely-operated underwater vehicles for ocean exploration

Language and Technical Expertise

Language proficiency: English (TOEFL-115/120), Hindi, Chinese (Basic), Japanese (Basic)

Programming languages: Python, C/C++, CUDA (basic), Matlab

Libraries: PyTorch, TensorFlow, Scipy, MegatronLM, DeepSpeed, ColossalAI

Hardware description : VHDL, Verilog, System Verilog, Perl, RDL, UPF

Other Presentations

- [1] STAR Lab, IIT Bombay, PS5 NSSS 2022, 1st February 2022 “ Development of a Compton Imaging Camera for Space Astrophysics” at the **21st National Space Science Symposium** (NSSS-2021), Astronomical Society of India (ASI)
- [2] Underwater Technology 2021 Online (UT21 Online) – **Underwater Research Video Competition**, Tokyo, Japan, 2 March, 2021 by IEEE Ocean Engineering Society. Young Researcher Award for Autonomous Underwater Vehicle Research Matsya by AUV-IITB
- [3] Poster presentation on “Reconfigurable Hardware Emulator of MEMS Gyroscopes with Built-in Error Source Models” at finals for **best student paper competition** at 2021 joint meeting of the IEEE European Frequency and Time Forum and International Frequency Control Symposium (**EFTF-IFCS**)-July 2021.

Research Projects

- Ongoing *Multi-host Distributed Shared Memory Systems using CXL*
A cache coherent interconnect to support remote memory sharing using Compute Express Link
- 2024-25 *MLCC - A Congestion Control technique for accelerating ML training*
A technique to address periodic DNN flow scheduling problem using congestion control algorithm
- 2020-21 *A re-configurable hardware emulator of MEMS gyroscopes*
An emulator model of MEMS vibratory gyroscopes on FPGA for real-time hardware-in-loop testing
- 2017-21 *MATSYA (AUV-IITB) : Autonomous Underwater Vehicle*
A high budget machine capable of performing naval missions with computer vision and acoustics
- 2020-21 *Modelling underwater channels for acoustic localization*
A software model of the non-ideal water channel for rapid testing of acoustic localization algorithms
- 2019 *DAKSHA - Detection of X-ray scattering events in space*
An x-ray capturing and imaging system deployed on satellites more efficient than ground processing
- 2021 *Random sampling techniques in high-dimensional spaces*
A survey of sampling arbitrary distributions in high-dimensional space using Markov chains
- 2020 *Supervised learning with a low precision representation of data*
A study of trade-off between the prediction accuracy and memory for training of ML classifiers

Positions of Responsibility

- 2020-21 **Team Leader - MATSYA (AUV-IITB)** *Multi-disciplinary Student Project Team*
- 2019-20 **Undergraduate Teaching Assistant** *Department Of Physics & Mathematics, IIT Bombay*

Extra-curricular

Sports :: Taekwondo, Yoga, Badminton, Swimming

Hobbies :: Dancing, Flute, Painting, Dry medium sketching, Bike racing